

Açai (*Euterpe oleracea* Mart.) polyphenolics in their glycoside and aglycone forms induce apoptosis of HL-60 leukemia cells.

The effects of açai polyphenolics on the antiproliferation and induction of apoptosis in HL-60 human leukemia cells were investigated. Interactions between anthocyanins and non-anthocyanin-polyphenolics in both their glycosidic and their aglycone forms were also investigated to determine additive or nonadditive responses. Polyphenolic fractions at 0.17-10.7 microM were found to reduce cell proliferation from 56 to 86% likely due to caspase-3 activation (apoptosis). Anthocyanin and polyphenolic fractions were nonadditive in their contribution to the cell antiproliferation activity. At equimolar concentrations, the glycosidic forms of phenolic acids and flavonoids induced a higher magnitude of change in cell parameters (proliferation and apoptosis) than their respective aglycone forms, while the opposite trend was observed for anthocyanin aglycones. This study demonstrated that açai offers a rich source of bioactive polyphenolics and confirmed the importance of investigating whole food systems when evaluating the potential health benefits of individual phytochemical compounds.